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I.D. # bakilić 2

Name Boran Aybak kilić

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Instructor

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Receiving or giving aid in a final examination is a cause for dismissal from the University.

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$$1) \quad \Psi(x, t=0) = \sqrt{\frac{2}{3a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{4}{3a}} \sin\left(\frac{5\pi x}{a}\right), \quad V(x) = \begin{cases} 0, & 0 \leq x \leq a \\ \infty, & \text{else} \end{cases}$$

$$1.a) \quad \Psi(x, t=0) = \frac{1}{\sqrt{3}} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{2}{3}} \sqrt{\frac{2}{a}} \sin\left(\frac{5\pi x}{a}\right) = \frac{1}{\sqrt{3}} \Psi_1(x) + \sqrt{\frac{2}{3}} \Psi_5(x)$$

5/5 where $\Psi_n(x)$ is the n -th energy eigenstate of ISW

$$\int_0^a |\Psi(x, t=0)|^2 dx = \frac{1}{3} \underbrace{\langle \Psi_1 | \Psi_1 \rangle}_{\text{normalized eistates}} + \frac{2}{3} \underbrace{\langle \Psi_5 | \Psi_5 \rangle}_{\text{normalized eistates}} + \frac{\sqrt{2}}{3} \underbrace{\langle \Psi_1 | \Psi_5 \rangle}_{\text{orthogonal states}} \\ = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 1 = 1, \Rightarrow \text{In fact, normalized } \checkmark$$

$$\Psi(x, t) = \frac{1}{\sqrt{3}} \Psi_1(x) e^{-iE_1 t/\hbar} + \sqrt{\frac{2}{3}} \Psi_5(x) e^{-iE_5 t/\hbar}, \quad \text{where } E_5 = 25E_1$$

$$\int_0^a |\Psi(x, t)|^2 dx = \frac{1}{3} \langle \Psi_1 | \Psi_1 \rangle + \frac{2}{3} \langle \Psi_5 | \Psi_5 \rangle + \frac{\sqrt{2}}{3} \langle \Psi_1 | \Psi_5 \rangle 2 \cos(24E_1 t/\hbar)$$

* Since cross-product term $\langle \Psi_1 | \Psi_5 \rangle = 0$ still due to orthogonality of eigenstates, $\int |\Psi(x, t)|^2 dx = 1$ always $\Rightarrow$ It's normalized at later times

$$1.b) \quad \langle H \rangle = \bar{E} = \sum_{n=1}^{\infty} P(E_n) E_n = \sum_{n=1}^{\infty} |c_n|^2 E_n \quad \text{where } \Psi(x) = \sum_{n=1}^{\infty} c_n \Psi_n(x)$$

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$$\langle H \rangle = \frac{1}{3} E_1 + \frac{2}{3} E_5 = \frac{51 E_1}{3} = \frac{51 \hbar^2 \pi^2}{6ma^2} \quad \checkmark$$

$\hookrightarrow$ This doesn't change with time since energy is conserved but one can also show that.

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$$

$$\begin{aligned} \langle \Psi(x,t) | H | \Psi(x,t) \rangle &= \left\langle \sqrt{\frac{1}{3}} \psi_1 e^{+iE_1 t/\hbar} + \sqrt{\frac{2}{3}} \psi_5 e^{-iE_5 t/\hbar} \left| \frac{\hbar^2}{2m} \left[\frac{1}{3} \left(\frac{\pi}{a} \right)^2 \psi_1 e^{-iE_1 t/\hbar} + \frac{4}{3} \left(\frac{5\pi}{a} \right)^2 \psi_5 e^{-iE_5 t/\hbar} \right] \right. \right\rangle \\ &= \frac{1}{3} E_1 + \frac{2}{3} E_5 + \frac{\sqrt{2}}{3} E_1 E_5 \underbrace{\langle \psi_1 | \psi_5 \rangle}_0 \cos(24 E_1 t/\hbar) \quad \checkmark \end{aligned}$$

1.c) Only energy values that can be observed are:

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$$\begin{aligned} E_1 &= \frac{\hbar^2 \pi^2}{2ma^2} \quad \text{with prob. } \frac{1}{3} \\ E_5 &= \frac{25\hbar^2 \pi^2}{2ma^2} \quad \text{with prob. } \frac{2}{3} \end{aligned} \quad \left. \begin{array}{l} \checkmark \\ \checkmark \end{array} \right\} \begin{array}{l} \text{They do not change} \\ \text{with time.} \end{array}$$

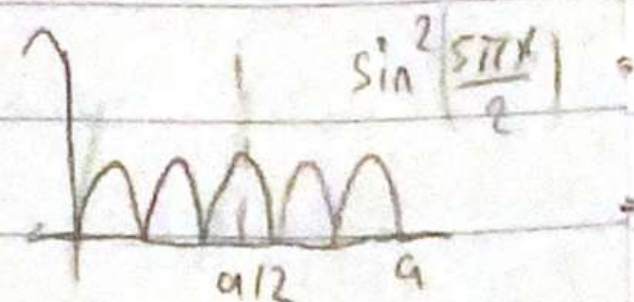
1.d) After observing E_5 , wavefunc will collapse to eigenstate ψ_5 with proper normalization:

$$\Psi(x, t > t_1) = \sqrt{\frac{2}{a}} \sin\left(\frac{5\pi x}{a}\right) e^{-iE_5 t/\hbar} \quad \checkmark$$

→ If the wavefunction doesn't decohere due to external noise, only E_5 can be observed at later times $t_2 > t_1$ ✓

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1.e) $\langle x \rangle = \frac{2}{a} \int_0^a x \sin^2\left(\frac{5\pi x}{a}\right) dx = \frac{a}{2}, \quad \checkmark$



→ due to $\sin^2\left(\frac{5\pi x}{a}\right)$ being symmetric around $a/2$

* This doesn't depend on time since complex exponential $e^{\pm iE_5 t/\hbar}$ cancel each other in $\Psi^*(x,t) \Psi(x,t)$. ✓

→

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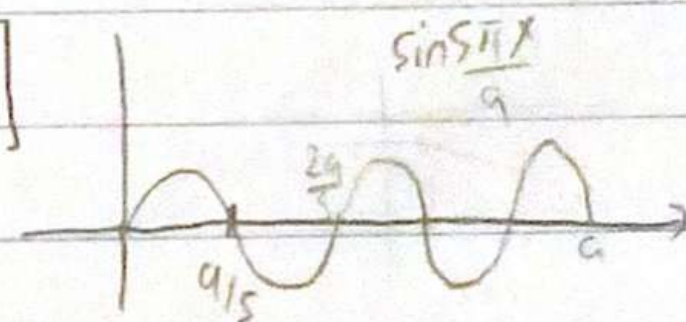
1.f) $\langle x \rangle_{\Psi(x,t=0)} = \int_0^a dx \Psi^*(x,t=0) x \Psi(x,t=0)$

$$= \frac{1}{3} \cdot \frac{2}{a} \int_0^a dx x \sin^2\left(\frac{\pi x}{a}\right) + \frac{2}{3} \cdot \frac{2}{a} \int_0^a dx x \sin^2\left(\frac{5\pi x}{a}\right) + \frac{\sqrt{2}}{3} \cdot \frac{2}{a} \cdot 2 \cos(24E_1 t/\hbar) \int_0^a dx x \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{5\pi x}{a}\right)$$

$$= \frac{1}{3} \cdot \frac{2}{a} \cdot \frac{a^2}{4} + \frac{2}{3} \cdot \frac{2}{a} \cdot \frac{a^2}{4} + \frac{2\sqrt{2}}{3} \cos(24E_1 t/\hbar) \int_0^a dx x \Psi_{11} \Psi_{51}$$

$$\frac{2}{a} \int_0^a dx \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{5\pi x}{a}\right) = \frac{1}{a} \int_0^a dx \left[\cos\left(\frac{4\pi x}{a}\right) - \cos\left(\frac{6\pi x}{a}\right) \right]$$

$$= \frac{1}{a} \left[\frac{a}{4\pi} \cos\left(\frac{4\pi x}{a}\right) - \frac{a}{6\pi} \cos\left(\frac{6\pi x}{a}\right) \right]_0^a = 0$$



$\langle x \rangle = \frac{a}{2}$ $\Rightarrow$ Doesn't depend on time since both wavefunctions are even in the cross product making integrand odd $\checkmark$

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1.g) In this case

$$\langle x \rangle = \frac{1}{3} \cdot \frac{a}{2} + \frac{2}{3} \cdot \frac{a}{2} + \frac{2\sqrt{2}}{3} \cos(15E_1 t/\hbar) \frac{2}{a} \int_0^a dx x \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{4\pi x}{a}\right)$$

even odd around $\frac{x}{2}$

$$= \frac{a}{2} + \frac{2\sqrt{2}}{3} \cos(15E_1 t/\hbar) \times \frac{1}{a} \int_0^a x \left[\cos\left(\frac{3\pi x}{a}\right) - \cos\left(\frac{5\pi x}{a}\right) \right] dx$$

$$2 \left[\left| \frac{a}{3\pi} \right|^2 - \left| \frac{a}{5\pi} \right|^2 \right] = \frac{32}{225} \frac{a^2}{\pi^2}$$

$$\langle x \rangle = \frac{a}{2} \left[1 + \frac{128}{675} \frac{a}{\pi^2} \cos\left(\frac{15 \hbar \pi^2}{2ma^2} t\right) \right] \Rightarrow \text{This time } \langle x \rangle \text{ depends on time since cross-terms integral doesn't cancel}$$

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$$2) \quad \Psi(x, t=0) = \int dk \frac{1}{\sqrt{2\pi}} e^{ikx} \tilde{\Psi}(k)$$

Just as in discrete case $\Psi(x, t) = \sum_n c_n \psi_n e^{-iE_n t/\hbar}$ we can write time-dependent sol as an integral of momentum eigenstates since each momentum eigstate is also an energy eigstate for free-particle

$$\Psi(x, t) = \int_{-\infty}^{\infty} dk \tilde{\Psi}(k) \frac{e^{ikx}}{\sqrt{2\pi}} e^{-iE(k)t/\hbar} \quad \checkmark$$

$$E(k) = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$\Psi(x, t) = \int_{-\infty}^{\infty} dk \frac{\tilde{\Psi}(k)}{\sqrt{2\pi}} e^{i(kx - \frac{\hbar k^2}{2m} t)}$$

For a good single particle wavefunc, it must be normalizable $\checkmark$

$$\begin{aligned} \int_{-\infty}^{\infty} dx |\Psi(x, t)|^2 &= \int dx \left[\tilde{\Psi}^*(k) \frac{e^{-ikx}}{\sqrt{2\pi}} e^{+i\frac{\hbar k^2}{2m} t} \right] \left[\tilde{\Psi}(k') \frac{e^{ik'x}}{\sqrt{2\pi}} e^{-i\frac{\hbar k'^2}{2m} t} \right] \\ &= \iint dk dk' \tilde{\Psi}^*(k) \tilde{\Psi}(k') e^{-i\hbar \frac{k^2 - k'^2}{2m} t} \underbrace{\int_{-\infty}^{\infty} dx \frac{e^{ikx}}{\sqrt{2\pi}} \frac{e^{ik'x}}{\sqrt{2\pi}}}_{\delta(k-k')} \end{aligned}$$

$$= \left[\int dk |\tilde{\Psi}(k)|^2 \right] = 1 \quad \checkmark$$

3) $\psi(x)$ with $\langle x \rangle_\psi$ and $\langle p \rangle_\psi$
 $\phi(x) = e^{ikx} \psi(x)$

3.a) $\langle \phi | \phi \rangle = \int_{-\infty}^{\infty} dx \phi^*(x) \phi(x) = \int_{-\infty}^{\infty} dx e^{-ikx} e^{ikx} \psi^*(x) \psi(x)$
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 $= \int_{-\infty}^{\infty} dx \psi^*(x) \psi(x) = \langle \psi | \psi \rangle \Rightarrow$ If $\langle \psi | \psi \rangle = 1$,
 then $\langle \phi | \phi \rangle = 1$

3.b) $\langle x \rangle_\phi = \int_{-\infty}^{\infty} dx e^{-ikx} \psi^*(x) x e^{ikx} \psi(x) = \int_{-\infty}^{\infty} dx \psi^*(x) x \psi(x) = \langle x \rangle_\psi$
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$$\langle p \rangle_\phi = \int_{-\infty}^{\infty} dx e^{-ikx} \psi^*(x) (-i\hbar) \frac{d}{dx} (e^{ikx} \psi(x))$$

$$= -i\hbar \int_{-\infty}^{\infty} dx e^{-ikx} \psi^*(x) (ike^{ikx} \psi(x) + e^{ikx} \frac{\partial \psi}{\partial x})$$

$$= \underbrace{\hbar k \int_{-\infty}^{\infty} dx \psi^*(x) \psi(x)}_{1 \text{ due to normalization}} + \int_{-\infty}^{\infty} dx \psi^*(x) (-i\hbar) \frac{\partial \psi}{\partial x}$$

$\hat{p} \psi(x)$

$$= \hbar k + \langle p \rangle_\psi$$

c) 0/5 * For some reason, I didn't see this question
 at all during exam

$$c) \quad \phi(x) = e^{ikx} \psi(x)$$

$$\text{Let us check } \hat{H} \phi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2 \phi(x)}{\partial x^2} + V(x) \phi(x)$$

$$\hat{H} \phi(x) = -\frac{\hbar^2}{2m} \frac{\partial}{\partial x} \left[i k e^{ikx} \psi(x) + e^{ikx} \frac{\partial \psi}{\partial x} \right] + V(x) e^{ikx} \psi(x)$$

$$= -\frac{\hbar^2}{2m} \left[-k^2 \psi(x) + 2ik \frac{\partial \psi}{\partial x} + \frac{\partial^2 \psi}{\partial x^2} \right] e^{ikx} + V(x) e^{ikx} \psi(x)$$

$$= e^{ikx} \left[\underbrace{-\frac{\hbar^2}{2m} \psi'' + V(x) \psi}_{\hat{H} \psi = E \psi} \right] + \frac{\hbar^2 k^2}{2m} e^{ikx} \psi(x) - \underbrace{\frac{i \hbar^2 k}{m} e^{ikx} \psi'(x)}_{-i \hbar \psi' = \hat{p} \psi}$$

$$= \left(E + \frac{\hbar^2 k^2}{2m} \right) \underbrace{e^{ikx} \psi(x)}_{\phi(x)} + \frac{\hbar k}{m} e^{ikx} \hat{p} \psi(x)$$

* For $\phi(x)$ to be an eigenstate, RHS must have been a constant multiple of $\phi(x)$. However there is an extra $\frac{\hbar k}{m} e^{ikx} \hat{p} \psi(x)$ term. Thus, $\phi(x)$ is NOT an eigenstate.